

ABDUL HANAN

AI/ML Engineer

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SUMMARY

AI/ML Engineer specializing in production RAG pipelines, LLM integration, multimodal retrieval systems, and MLOps deployment. Experienced designing end-to-end AI workflows—from embedding generation and vector database architecture (FAISS, pgvector, Neo4j) to LangChain-orchestrated LLM applications on AWS. Proven ability to ship intelligent, scalable AI systems across clinical AI, migration, and e-commerce recommendation domains.

EDUCATION

FAST National University of Computer & Emerging Sciences

B.S. in Artificial Intelligence | August 2022 – May 2026

EXPERIENCE

AI/ML Engineer — Contract

Cognara Training | November 2025 – April 2026 | LangChain, RAG, pgvector, FastAPI, Docker, AWS

- Designed and deployed production AI systems—APD Listener Tool and Tradie Migration Platform—implementing RAG pipelines, LLM integrations, and AI-powered API services end-to-end.
- Built LangChain + pgvector RAG pipeline with OpenAI embeddings for semantic document retrieval and intelligent candidate-visa case matching; deployed on AWS EC2 via Docker/Nginx.
- Engineered multithreaded queue-based audio-to-text AI pipeline on AWS EC2/S3 for high-concurrency processing with sub-second retrieval latency.
- Taught international students Python from beginner to advanced level, covering OOP, automation, REST APIs, web scraping, backend development, and applied AI/ML concepts.

AI Engineer Intern

One Tech and AI | July 2025 – September 2025 | RAG, LLMs, Docker, FastAPI

- Built and deployed RAG-based conversational AI agents with full ownership of retrieval pipeline architecture, LLM orchestration, and containerized production API services.
- Conducted applied Quantum AI research, prototyping hybrid quantum-classical ML experiments over 3 months.

SKILLS & TOOLS

AI & ML: LangChain, RAG pipelines, OpenAI/MedGemma APIs, Hugging Face Transformers, CLIP, MiniLM, PyTorch, TensorFlow

Vector & Search: FAISS, pgvector, Annoy, Neo4j knowledge graphs, semantic embedding pipelines, Qiskit

MLOps & Infra: Docker, AWS (EC2, S3, SageMaker, Lambda, IAM), MLflow, Apache Airflow, DVC, GitHub Actions CI/CD

Backend & Data: Python, FastAPI, PostgreSQL, SQL, REST APIs, Nginx, Cloudflare Tunnel

SELECTED PROJECTS

Doctor Path AI — Clinical AI Decision Support

pgvector, Neo4j, FAISS, Docker, FastAPI, MedGemma, LangChain | Aug 2025 – Feb 2026

- Production clinical AI on MIMIC-III: multimodal text+image LLM pipelines, hybrid retrieval architecture (pgvector embeddings + Neo4j clinical knowledge graphs + FAISS case similarity ranking).
- RAG-based evidence pipeline surfacing historically similar cases for context-aware treatment recommendations; containerized with Docker and exposed via FastAPI.

Tradie Migration Platform — RAG-Powered SaaS

FastAPI, LangChain, OpenAI, pgvector, PostgreSQL, Docker | Mar – Apr 2026

- LangChain + pgvector RAG pipeline with OpenAI embeddings for semantic visa document retrieval and AI-powered candidate matching across complex migration workflows.
- Secure FastAPI backend (JWT RBAC, PostgreSQL) with automated OTP/email workflows supporting end-to-end AI case management at production scale.

Hybrid E-Commerce Recommendation System

FAISS, Annoy, CLIP, PyTorch, MiniLM | Mar – May 2025

- Multimodal recommendation engine fusing MiniLM text + CLIP image embeddings for cross-modal semantic product discovery.
- Two-stage ANN retrieval: Annoy projection-tree coarse filtering → FAISS IVF fine-grained re-ranking; benchmarked recall-latency trade-offs across million-item catalogs.

APD Listener Tool — Audio AI Pipeline

FastAPI, Docker, AWS EC2/S3, ASR/NLP, Nginx | Nov 2025 – Feb 2026

- End-to-end audio AI pipeline (ASR + NLP post-processing) converting podcasts to structured instructions; multithreaded queue backend on AWS EC2/S3 for concurrent high-throughput jobs.

HONORS & ACTIVITIES

- Top 50 Nationally — ICPC 2026 | Head of Job Fair 2026, Co-Head 2025, Coordinator 2024 — FAST NU
- Teaching Assistant: Generative AI, Deep Learning, DB Systems, Assembly Language — FAST NU CFD | 350+ LeetCode
- Python Instructor — Taught international students Python (beginner to advanced): OOP, automation, REST APIs, web scraping, backend development, and AI/ML concepts | July 2025 – January 2026
- Level 2 Fiverr Freelancer — AI Automation & ML Engineering